Milestone 4

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DSC640: Data Presentation & Visualization

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Final Project Report: The National Childcare Affordability Crisis (2008–2018)

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Target Audience: State Legislators and Labor Departments

Summary of Analysis

This project investigates the economic sustainability of childcare in the United States using the National Database of Childcare Prices (NDCP). Spanning a decade (2008–2018) and encompassing over 34,000 records, the analysis identifies the "Affordability Gap"—the disparity between the market price of care and the federal Department of Health and Human Services (HHS) benchmark, which suggests childcare should not exceed 7% of a family's income. By processing Median Household Income (MHI) against annualized costs for various age groups, this study provides a longitudinal look at how childcare costs act as a macroeconomic anchor.

Findings

The analysis reveals a systemic failure to meet federal affordability standards. On average, infant center-based care requires **16.39%** of a median household's income, more than double the HHS recommendation. While the national average is concerning, the regional data reveals a crisis: in high-burden "hotspots," the cost of care peaks at **54.3%** of gross income. Furthermore, while wages saw a stagnant 3.2% growth over the study period, preschool care costs escalated by over 11%, illustrating a "scissors effect" where prices are rising significantly faster than the ability to pay.

Assumptions & Items for Clarification

This analysis assumes that the weekly prices provided represent full-time care for a single child. Additionally, it assumes that Median Household Income is a sufficient proxy for a family's ability to pay, though it does not account for existing debt or local cost-of-living variances. One item requiring further clarification is the exact nature of the "imputed" data points (prefixed with "i" in the source). While used to provide a comprehensive national view, a deeper understanding of the imputation methodology would strengthen the precision of regional variance reporting.

Direction of Story & Target Audience

The narrative, titled **"The 16% Reality,"** is directed at State Legislators and Labor Departments. The goal is to move the conversation from "personal family responsibility" to "economic infrastructure." By showing that childcare is mathematically impossible for many median-wage earners, the story advocates for localized subsidy reform and a re-evaluation of the 7% federal benchmark in the face of modern market realities.

Mediums Included & Design Decisions

For the final presentation, I developed an **Interactive Python Dashboard** using Dash and Plotly.

- **Why:** Unlike a static PDF, an interactive dashboard allows legislators to filter by their specific state and year, making the data personally relevant to their constituents.
- **Design Decisions:** I implemented a **"Red Alert" color palette** (Orange-to-Red sequential scale). Dark red is used strictly for counties where the burden exceeds 15% (double the goal). Sizing and spacing were optimized to ensure the **Year Scrubber** remained the focal point, allowing the audience to watch the "reddening" of the U.S. map over the decade as affordability collapsed.

Ethical Considerations

- **Data Transformation:** Data was transformed from weekly prices to "Affordability Ratios" (Annual Cost / MHI). While this makes the data more relatable, it risks oversimplifying complex financial lives.
- **Legal & Regulatory Guidelines:** The data is sourced from the U.S. Department of Labor and is public domain. There are no HIPAA or PII (Personally Identifiable Information) concerns, as the data is aggregated at the county level.
- **Risk of Presentation:** The primary risk is the "Average Trap." By presenting a 16.39% average, we may hide the struggle of those in the 54.3% burden zones if not clearly labeled. I mitigated this by including a "Leaderboard" to highlight extremes.
- **Sourcing & Credibility:** The data is a national database verified by the Women's Bureau. It was acquired ethically via federal open-data portals.
- **Assumptions in Cleaning:** No data was filtered out without labeling. Missing values in certain years were handled via Pandas to ensure the longitudinal mean remained accurate without "hallucinating" values for missing counties.

Lessons Learned

What would I do differently? Next time, I would incorporate cost-of-living indices (housing and transportation) alongside childcare. Childcare does not exist in a vacuum, and seeing how it competes with rent would tell an even more harrowing story of the modern family budget.

What did I enjoy the most? Developing the interactive dashboard in Python was the most rewarding aspect. Transitioning from a static spreadsheet to a "living" tool where you can drag a slider and see a decade of economic history change in real-time was a powerful reminder of how data visualization can turn dry statistics into a compelling, human story.